Oluwasola Oyinloye

Geospatial data scientist · Geospatial AI specialist

GIS · Remote sensing · Drone mapping · Geospatial AI

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I'm a geospatial data scientist and geospatial AI specialist in Lagos with five years of professional GIS and remote sensing work. I plan and fly drone surveys in Nigeria and work remotely from satellite data for clients across Africa and beyond. Every result I deliver comes with how it was validated and the code that produced it.

SERVICES

- **Geospatial AI and machine learning.** Deep-learning models that find and count objects in imagery, and predictive models tested on ground they have not seen.
- **Drone mapping and photogrammetry.** I plan and fly the survey, then turn the imagery into orthomosaics, surface and terrain models, and counts.
- **Satellite remote sensing.** Land cover, vegetation, water and change over time, read from optical and radar satellite imagery, anywhere.
- **GIS mapping and spatial analysis.** Site selection, access and proximity analysis, map production and clean, well-structured spatial data.
- **Flood, climate and environmental risk.** Where flooding is more likely, which people, roads and facilities sit in the way, and how land cover is changing.
- **Web GIS development and dashboards.** Interactive web maps, Streamlit apps and Power BI dashboards that work on any device.
- **Spatial databases and data engineering.** PostGIS databases and repeatable Python and Earth Engine pipelines that keep spatial data current.
- **Training and capacity building.** Hands-on Python, GIS and remote sensing training for teams, from small workshops to a 1,500-person seminar.

SELECTED WORK

Oil palm counting from aerial imagery

For a commercial oil palm plantation: a YOLOv8 crown-detection pipeline that mapped 718,751 palms across 12,166.6 ha in 26 blocks. Overlapping deliveries were resolved so every hectare is counted once, removing 302,412 double counts.

GeoAID flood susceptibility, Amuwo-Odofin (Lagos)

An open model with a live dashboard: structural flood susceptibility, near-real-time rainfall and exposure of people, schools, clinics and roads. ROC-AUC about 0.798 on a random split and about 0.642 under spatial cross-validation, reported side by side.

Lagos Access Map (in development)

A ward-by-ward audit of how complete OpenStreetMap's record of health facilities is in Lagos, with confidence intervals, as the base for access-to-care mapping.

SECTORS

- Agriculture and plantations
- Agric lending and finance
- Real estate, construction and infrastructure
- Government and urban planning
- NGOs, health and development
- Environment, climate and insurance
- Mining, quarrying and energy

WHERE

On site in Nigeria. Remote across Africa and worldwide.

HOW I WORK

1. A 20-minute scoping call
2. A pilot on your data, with its error rate
3. Delivery with code and documentation you keep

TOOLS

Programming and analysis: Python, SQL, Jupyter, pandas, NumPy, SciPy, statsmodels. **Machine learning and AI:** PyTorch, Ultralytics YOLOv8, scikit-learn, XGBoost, SHAP, OpenCV. **Earth observation:** Google Earth Engine, geemap, Sentinel-1 SAR, Sentinel-2, SRTM, CHIRPS, NASA GPM IMERG, ESA WorldCover, WorldPop. **GIS and spatial data:** QGIS, Google Earth Pro, PostGIS, GeoPandas, Rasterio, Shapely, PyProj, OpenStreetMap. **Drone mapping:** DJI Terra, Agisoft Metashape, WebODM / OpenDroneMap, GNSS ground control. **Dashboards and web maps:** Streamlit, Power BI, Plotly, Folium, Leaflet.

EDUCATION

- Master of Information Technology (MIT), AI specialisation, Miva Open University
- BSc Surveying and Geoinformatics, University of Ilorin

TEACHING

Taught Python for geospatial analysis to a seminar of 1,500 people.